## Chapter C6: Making useful chemicals

**Overview**

This chapter considers the laboratory and large-scale production of useful chemicals.

Topic C6.1 begins with the laboratory synthesis of salts from acid reactions, and also looks at the characteristics of both acids and bases.

In Topic C6.2, the story moves on to study how chemists manage the rate of reaction when these reactions take place, in the context of managing conditions both in the laboratory and in industry. This topic gives the opportunity for a wide range of practical investigation and mathematical analysis of rates.

Topic C6.3 looks at reversible reactions, with particular emphasis on the large-scale production of ammonia.

Topic C6.4 develops all the ideas in the chapter together to look at both how the conditions and 'routes' for a chemical process are chosen by thinking about the economic and environmental issues surrounding the production of chemicals on a large scale. The manufacture of fertilisers is used as a context for considering how chemists reach decisions about the optimum processes for large- scale production of bulk chemicals.

## Learning about Making useful Chemicals before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- explain that some changes result in the formation of new materials, and that this kind of change is not usually reversible
- represent chemical reactions using formulae and using equations
- define acids and alkalis in terms of neutralisation reactions
- describe the pH scale for measuring acidity/alkalinity; and indicators
- recall reactions of acids with metals to produce a salt plus hydrogen and reactions of acids with alkalis to produce a salt plus water
- know what catalysts do.
- know about energy changes on changes of state (qualitative)
- know about exothermic and endothermic chemical reactions (qualitative).

## Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.